Lichens - *Anabaena* and *Nostoc*
*Anthoceros* - *Nostoc*
*Azolla* - *Anabaena azollae*
*Cycas* - *Anabaena* and *Nostoc*

### b. Non-symbiotic Nitrogen fixation

Free living bacteria and fungi also fix atmospheric nitrogen.

| | |
|---|---|
| Aerobic | *Azotobacter, Beijerinckia and Derxia* |
| Anaerobic | *Clostridium* |
| Photosynthetic | *Chlorobium and Rhodospirillum* |
| Chemosynthetic | *Disulfovibrio* |
| Free living fungi | Yeast and *Pullularia* |
| Cyanobacteria | *Nostoc, Anabaena and Oscillatoria.* |

## 12.8 Nitrogen cycle and nitrogen metabolism

### 12.8.1 Nitrogen cycle

This cycle consists of following stages:

### 1. Fixation of atmospheric nitrogen

Di-nitrogen molecule from the atmosphere progressively gets reduced by addition of a pair of hydrogen atoms. Triple bond between two nitrogen atoms (N≡N) are cleaved to produce ammonia (Figure 12.7).

Nitrogen fixation process requires Nitrogenase enzyme complex, Minerals (Mo, Fe and S), anaerobic condition, ATP, electron and glucose 6 phosphate as $H^+$ donor. Nitrogenase enzyme is active only in anaerobic condition. To create this anaerobic condition a pigment known as **leghaemoglobin** is synthesized in the nodules which acts as oxygen scavenger and removes the oxygen. Nitrogen fixing bacteria in root nodules appears pinkish due to the presence of this leghaemoglobin pigment.

Overall equation:

$$N_2 + 8e^- + 8H^+ + 16ATP \longrightarrow 2NH_3^+ + H_2 + 16ADP + 16\ Pi$$

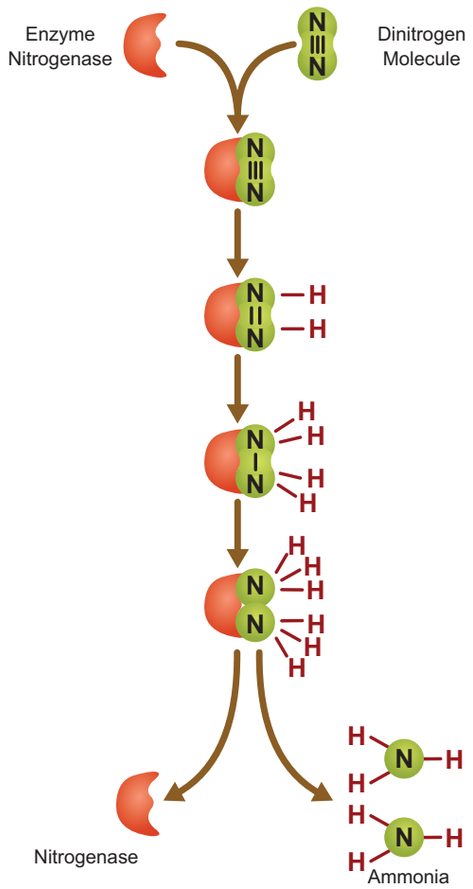



**Figure 12.7:** Nitrogenase enzyme function

### 2. Nitrification

Ammonia ($NH_3^+$) is converted into Nitrite ($NO_2^-$) by *Nitrosomonas* bacterium. Nitrite is then converted into Nitrate ($NO_3^-$) by *Nitrobacter* bacterium.

Plants are more adapted to absorb nitrate ($NO_3^-$) than ammonium ions from the soil.

$$2\ NH_3^+ + 3\ O_2 \xrightarrow{Nitrosomonas} 2\ NO_2^- + 2\ H^+ + 2H_2O$$

$$2\ NO_2^- + O_2 \xrightarrow{Nitrobacter} 2\ NO_3^-$$

### 3. Nitrate Assimilation

The process by which nitrate is reduced to ammonia is called **nitrate assimilation** and occurs during nitrogen cycle.

$$NO_3^- \xrightarrow[Mo]{\text{Nitrate reductase}} NO_2^-$$

$$NO_2^- \xrightarrow[Cu,\ Fe]{\text{Nitrite reductase}} NH_3^+$$



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